Taipei Veterans General Hospital

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Sample Information

Patient Name: 朱森堯
Gender: Male
ID No.: A124992382
History No.: 40887318
Age: 42

Ordering Doctor: DOC8716K 李怡緻
Ordering REQ.: 0BNMFRH
Signing in Date: 2021/11/18

Path No.: S110-89554
MP No.: F21096
Assay: Oncomine Focus Assay
Sample Type: FFPE
Block No.: S110-04165C
Percentage of tumor cells: 40%

Reporting Doctor: DOC5466K 葉奕成 (Phone: 8#5466)

Note:

Sample Cancer Type: Gastric Cancer

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Report Highlights

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Relevant Gastric Cancer Variants

Gene	Finding
ERBB2	None detected
NTRK1	None detected
NTRK2	None detected
NTRK3	None detected

Relevant Biomarkers

No relevant biomarkers found in this sample.

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Variant Details

DNA Sequence Variants

Gene	Amino Acid Change	Coding	Variant ID	Locus	Allele Frequency	Transcript	Variant Effect	Coverage
JAK1	p.(P733=)	c.2199A>G	.	chr1:65310489	43.70%	NM_002227.4	synonymous	2000
FGFR4	p.(P136L)	c.407C>T	.	chr5:176517797	99.64%	NM_213647.3	missense	1967
RET	p.(L769=)	c.2307G>T	.	chr10:43613843	100.00%	NM_020975.6	synonymous	1999

Copy Number Variations

Gene	Locus	Copy Number
FGFR2	chr10:123247505	78.93

Biomarker Descriptions

FGFR2 (fibroblast growth factor receptor 2)

Background: The FGFR2 gene encodes fibroblast growth receptor 2, a member of the fibroblast growth- factor receptor (FGFR) family that also includes FGFR1, 3, and 4. These proteins are single-transmembrane receptors composed of three extracellular immunoglobulin (Ig)-type domains and an intracellular kinase domain. Upon FGF-mediated stimulation, FGFRs activate several oncogenic signaling pathways, including the RAS/RAF/MEK/ERK, PI3K/AKT/MTOR, PLC/PKC, and JAK/STAT pathways influencing cell proliferation, migration, and survival[1,2,3].

Alterations and prevalence: Aberrations most common to the FGFR family are amplifications, followed by mutations and fusions. The majority of these aberrations result in gain of function[4]. Missense mutations are the most prevalent alterations in FGFR2 and are observed in up to 15% of uterine carcinomas[5,6,7]. These mutations are predominantly activating, most often involve substitutions at S252 and P253, and confer sensitivity to pan-FGFR2 inhibitors[5,8]. FGFR2 amplification occurs in up to 4% of gastric carcinoma, and is associated with poor prognosis as well as tumor invasion and metastasis[6,9,10,11]. FGFR2 fusions have also been reported in up to 14% of cholangiocarcinoma and confer sensitivity to select FGFR inhibitors[6,12,13].

Potential relevance: The pan-FGFR inhibitor, infigratinib, has been granted accelerated approval (2021) for previously treated, unresectable locally advanced or metastatic cholangiocarcinoma positive for FGFR2 fusion or other rearrangement[14]. The pan-FGFR inhibitor, erdafitinib[15], received FDA approval (2019) for the treatment of locally advanced or metastatic urothelial cancer that is positive for FGFR2 fusions including, FGFR2-BICC1 and FGFR2-CASP7, FGFR3 fusions, or FGFR3 mutation. Additionally, the Pan-FGFR inhibitor, futibatinib[16], has been granted Breakthrough Therapy Designation(2021) for FGFR2 rearrangement or fusion-positive locally advanced or metastatic cholangiocarcinoma. The FGFR kinase inhibitor, pemigatinib[17], received FDA approval (2020), for previously treated, advanced or unresectable cholangiocarcinoma harboring FGFR2 fusions or other FGFR2 rearrangements. The FDA also granted fast-track designation (2018) to Debio 1347[18] for solid tumors harboring FGFR1, FGFR2, or FGFR3 aberrations. Additional FGFR inhibitors are under clinical evaluation for FGFR2 aberrations. In a phase II study of patients with FGFR2 fusion-positive intrahepatic cholangiocarcinoma, the pan-kinase inhibitor derazantinib, demonstrated an overall response rate (ORR) of 20.7% with progression-free survival (PFS) of 5.7 months[19]. Likewise, results of a phase II trial testing the pan-FGFR inhibitor, infigratinib (BGJ398) demonstrated an ORR of 14.8% (18.8% FGFR2 fusions only), disease control rate (DCR) of 75.4% (83.3% FGFR2 fusions only), and a median PFS of 5.8 months[20].

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Clinical Trials in Taiwan region:

Alerts Informed By Public Data Sources

Current FDA Information

Contraindicated Not recommended Resistance Breakthrough Fast Track

FDA information is current as of 2021-08-18. For the most up-to-date information, search www.fda.gov.

FGFR2 amplification

Debio 1347

Cancer type: Solid Tumor **Variant class:** FGFR2 aberration

Supporting Statement:
The FDA has granted Fast Track Designation to the FGFR 1-3 inhibitor, debio 1347, for FGFR1/2/3 alterations in unresectable or metastatic solid tumors.

Reference:

https://www.debiopharm.com/drug-development/press-releases/fda-grants-fast-track-designation-to-debiopharm-internationals-debio-1347-for-the-treatment-of-patients-with-unresectable-or-metastatic-tumors-with-a-specific-fgfr-gene-alteration/

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Signatures

Testing Personnel: Shin-Ying Lin 2021.11.19 15:50:10 +08'00'

Laboratory Supervisor: 2021.11.19 15:50:23 +08'00'

Pathologist: Yi-Chu Yeh 2021.11.19 15:50:35 +08'00'

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